

## Multiple Choice

1. **Answer:** D

*Options:* A) 6; B) 3; C) All the terms are odd; D) 4; E) 1

*Note:* Correct option: D - 4

2. **Answer:** D

*Options:* A)  $f(d) > f(c)$ ; B)  $f(c) < f(d)$ ; C)  $f(d) \neq f(c)$ ; D)  $f(c) = f(d)$ ; E)  $f(d) < f(c)$

*Note:* Correct option: D -  $f(c) = f(d)$

3. **Answer:** A

*Options:* A) Statement 1 is always true, Statement 2 is false only for certain groups; B) True, True; C) Statement 1 is always false, Statement 2 is true only for certain groups; D) True, False; E) Statement 1 is true only for certain groups, Statement 2 is always true

*Note:* Correct option: A - Statement 1 is always true, Statement 2 is false only for certain groups

4. **Answer:** A

*Options:* A) 42; B) 30; C) -42; D) -36; E) 1 over 216

*Note:* Correct option: A - 42

5. **Answer:** A

*Options:* A) 30; B) 15; C) 12; D) 6; E) 24

*Note:* Correct option: A - 30

## True / False

6. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: '80'.

7. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: '2'.

8. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: 'Yes, with  $p=3$ '.

9. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: '8'.

10. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: '94'.

## Long Form Response

11. **Answer:** Dividing the equation by  $x^2$ , we get

$$6x^2 - 35x + 62 - \frac{35}{x} + \frac{6}{x^2} = 0.$$

Let  $y = x + \frac{1}{x}$ . Then

$$y^2 = x^2 + 2 + \frac{1}{x^2},$$

so  $x^2 + \frac{1}{x^2} = y^2 - 2$ . Thus, we can re-write the equation above as

$$6(y^2 - 2) - 35y + 62 = 0.$$

This simplifies to  $6y^2 - 35y + 50 = 0$ . The roots are  $y = \frac{5}{2}$  and  $y = \frac{10}{3}$ . The roots to

$$x + \frac{1}{x} = \frac{5}{2}$$

are 2 and  $\frac{1}{2}$ . The roots to

$$x + \frac{1}{x} = \frac{10}{3}$$

are 3 and  $\frac{1}{3}$ . Thus, the roots of  $6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0$  are 2, 3,  $\frac{1}{2}$ ,  $\frac{1}{3}$ .

*Note:* Students should show all steps in their mathematical solution.

12. **Answer:** Leslie spent 20 dollars and 8 cents at a store. How is the amount of money Leslie spent written as a fraction?. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

*End of Answer Key*